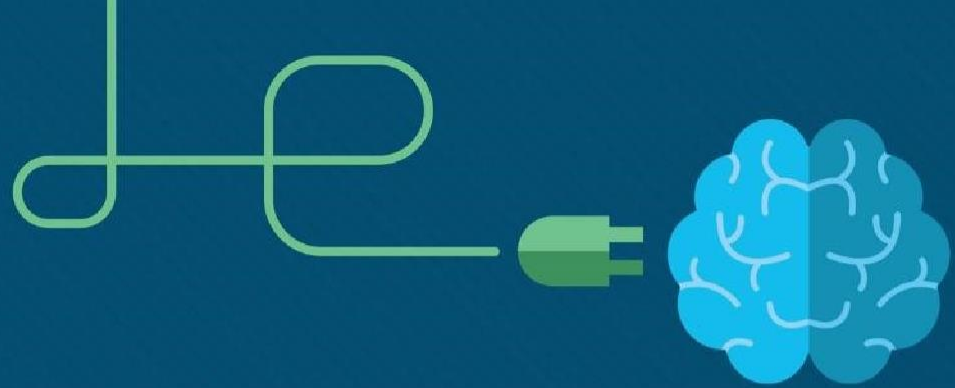




Chapter 2: Scaling VLANs

CCNA Routing and Switching

Scaling Networks v6.0



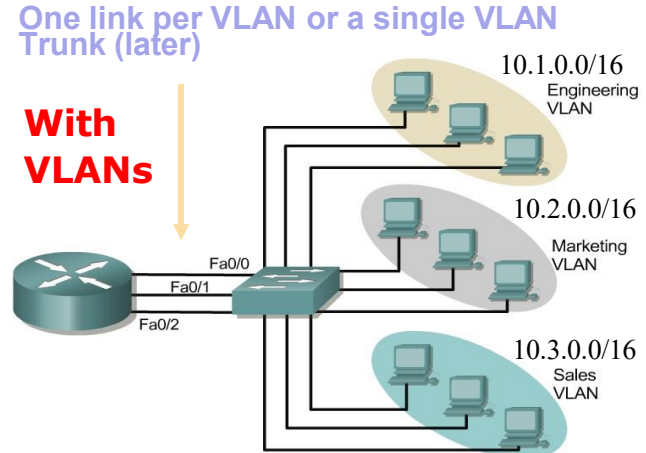
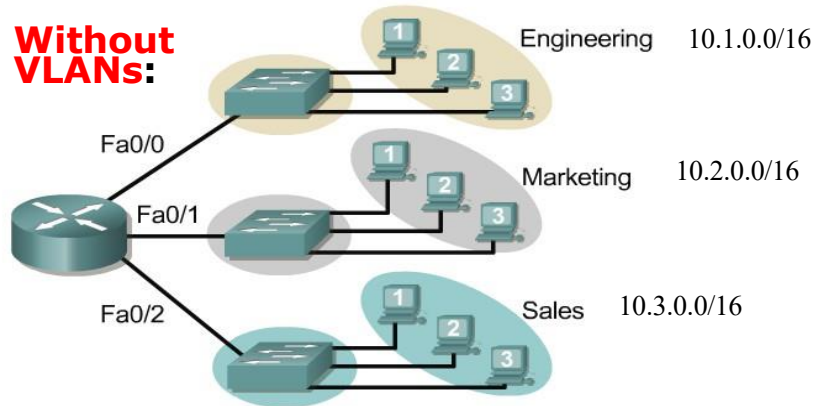
CCNA



CCNA Course Flow

Broadcast domains with VLANs and routers

- Without VLANs, each group is on a different IP network and on a different switch.
- Using VLANs. Switch is configured with the ports on the appropriate VLAN. Still, each group on a different IP network; however, they are all on the same switch.
- What are the broadcast domains in each?

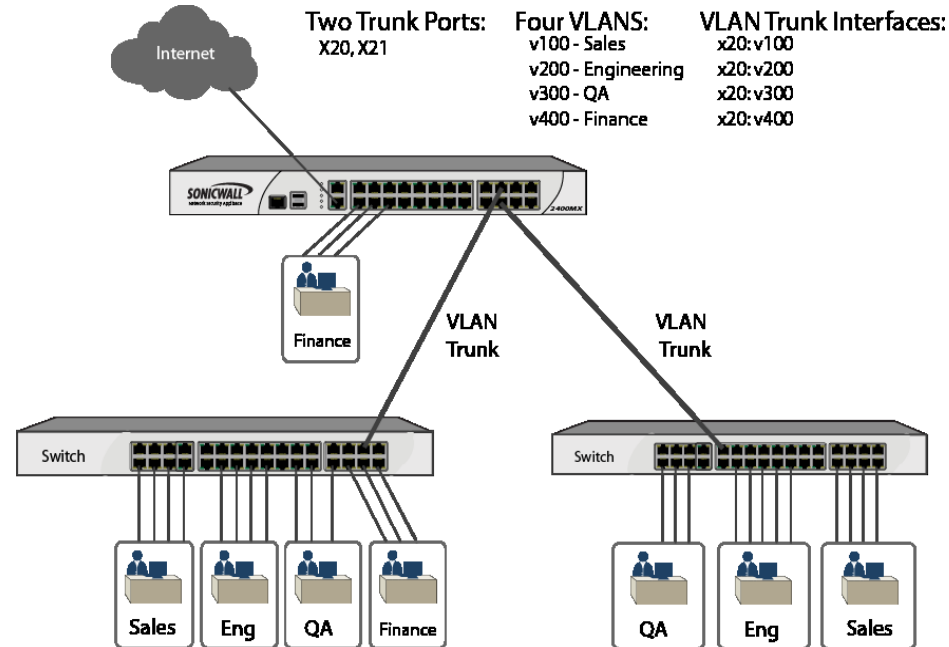


Local VLANs

- 2 VLANs or more within a single switch
- VLANs address scalability, security, and network management.
- Routers in VLAN topologies provide broadcast filtering, security.
- **Edge ports**, where end nodes are connected, are configured as members of a VLAN
- The switch behaves as several virtual switches, sending traffic only within VLAN members.
- **Switches** may not bridge any traffic between VLANs.
- Traffic should only be routed between VLANs.

VLANs across switches

- Two switches can exchange traffic from one or more VLANs
- Inter-switch links are configured as **trunks**, carrying frames from all or a subset of a switch's VLANs
- **Trunking** is the point to point connection between more than one Ethernet switch and some other network devices like switch or a router.
- Each frame carries a **tag** that identifies which VLAN it belongs to



2.1 VTP, Extended VLANs, and DTP

VTP Overview

When a new **VLAN is created** and configured on a switch **without the VTP protocol** enabled, this must be **manually replicated to all switches on the network** so they are all aware of the newly created VLAN. This means that the administrator must configure each switch separately, a task that requires a lot of time and adds a considerable amount of overhead depending on the size of the network.

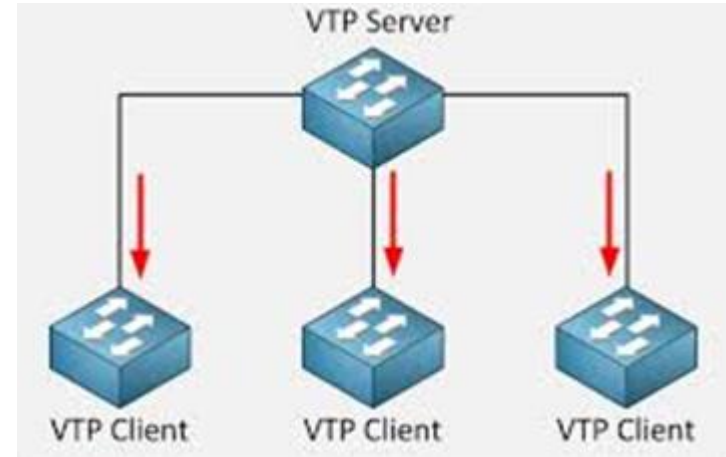
The configuration of a VLAN includes the VLAN number, name and a few more parameters which will be analysed further on. This information is then stored on each switch's NVRAM and any VLAN changes made to any switch must again be replicated manually on all switches.

- **With the VTP protocol** configured and operating, you can forget about running around making sure you have updated all switches as you only need to make the **changes** on the nominated **VTP server switch(es)** on your network. This will also ensure these changes are magically propagated to all other switches regardless of where they are.

VTP Concepts and Operation

VTP Overview

- As the number of switches increases on a small or medium-sized business network, the overall administration required to manage virtual local area networks (VLANs) and trunks in the network becomes challenging.
- **What?** VLAN Trunking Protocol (VTP) is protocol used in scalable VLAN to reduces administration in a switched network.
- **How?** The VTP server distributes and synchronizes VLAN information over trunk links to VTP- enabled switches throughout the switched network.
- **Why?** This eliminates the need to configure the new VLAN on every switch.
- **VTP** is a Cisco proprietary protocol that is available on most of the **Cisco Catalyst Series products.**



VTP Overview

- VLAN trunking protocol (VTP) allows a network administrator to manage VLANs on a switch configured as a VTP server.
- The VTP server distributes and synchronizes VLAN information over trunk links to VTP-enabled switches throughout the switched network.

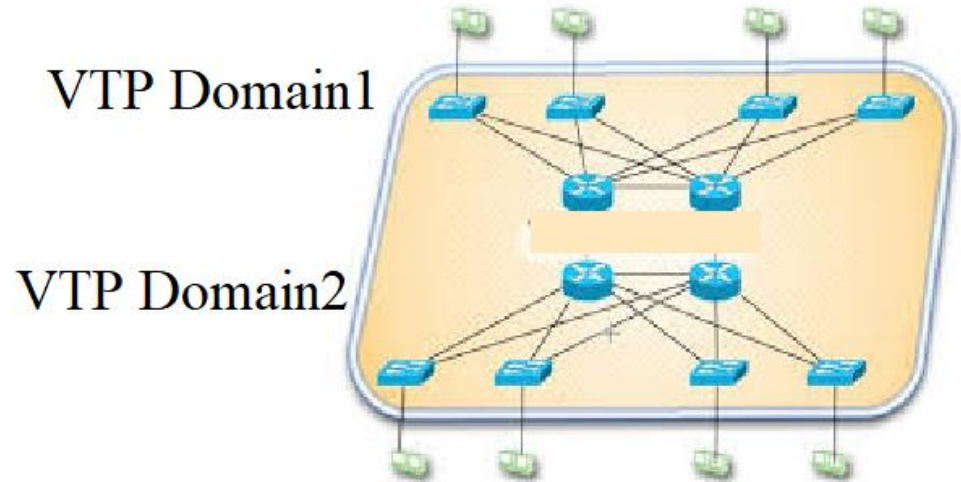
VTP Components	Definition
VTP Domain	▪ Consists of one or more interconnected switches. ▪ All switches in a domain share VLAN configuration details using VTP advertisements. ▪ Switches that are in different VTP domains do not exchange VTP messages. ▪ A router or Layer 3 switch defines the boundary of each domain.
VTP Advertisements	▪ Each switch in the VTP domain sends periodic global configuration advertisements from each trunk port to a reserved multicast address. ▪ Neighboring switches receive these advertisements and update their VTP and VLAN configurations as necessary.
VTP Modes	A switch can be configured in one of three VTP modes: server, client, or transparent.
VTP Password	Switches in the VTP domain can be also be configured with a password.

Note: VTP advertisements will not be exchanged if the trunk between the switches is inactive or if the trunk is misconfigured.

VTP Concepts and Operation

VTP Domain

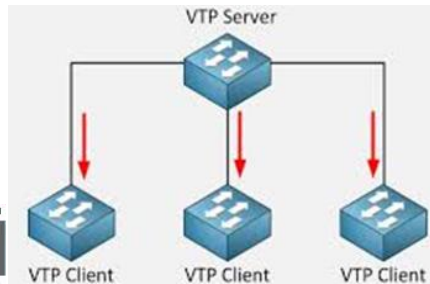
- Consists one or more interconnected switches.
- All switches in domain share VLAN configuration using VTP advertisements
- Switches in different VTP domains don't exchange VTP messages.
- Router define the boundary in each domain



VTP Concepts and Operation

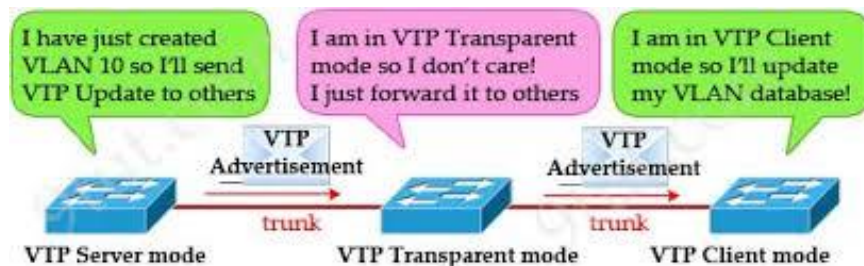
VTP Modes

VTP Mode	Definition
VTP Server	• VTP servers advertise the VTP domain VLAN information to other VTP-enabled switches in the same VTP domain. • VTP servers store the VLAN information for the entire domain in NVRAM. • Switches configured in VTP server mode are allowed to create, delete, or rename VLANs for the domain.
VTP Client	• VTP clients function the same way as VTP servers, but you cannot create, change, or delete VLANs on a VTP client. • A VTP client only stores the VLAN information for the entire domain while the switch is on. • A switch reset deletes the VLAN information. • You must configure VTP client mode on a switch.
VTP Transparent	• Transparent switches do not participate in VTP except to forward VTP advertisements to VTP clients and VTP servers. • VLANs that are created, renamed, or deleted on transparent switches are local to that switch only. • To create an extended VLAN, a switch must be configured as a VTP transparent switch when using VTP versions 1 or 2.



VTP Concepts and Operation

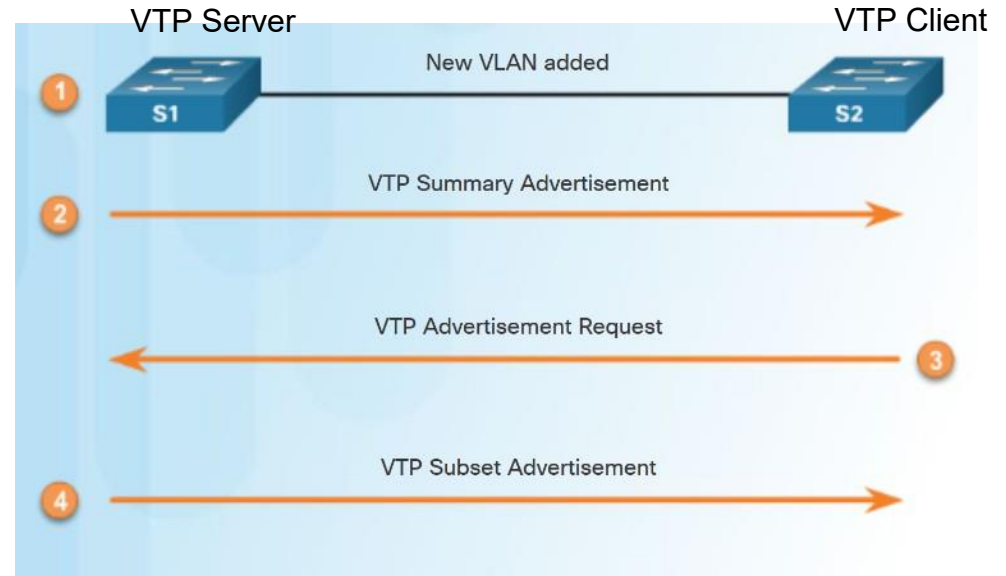
VTP Question	VTP Server	VTP Client	VTP Transparent
What are the differences?	• Manages domain and VLAN configuration. • Multiple VTP servers can be configured.	• Updates local VTP configurations. • VTP client switches cannot change VLAN configurations.	• Manages local VLAN configurations. • VLAN configurations are not shared with VTP network.
Does it respond to VTP advertisements?	Participates fully	Participates fully	Only forwards VTP advertisements
Is the global VLAN configuration preserved on restart?	Yes, global configurations are stored in NVRAM	No, global configurations are stored in RAM only	No, local VLAN configuration is only stored in NVRAM
Does it update other VTP-enabled switches?	Yes	Yes	No



VTP Concepts and Operation

VTP Advertisements

VTP Advertisements · **VTP uses various types of messages or advertisements** to allow VTP enabled switches to exchange VLAN information.

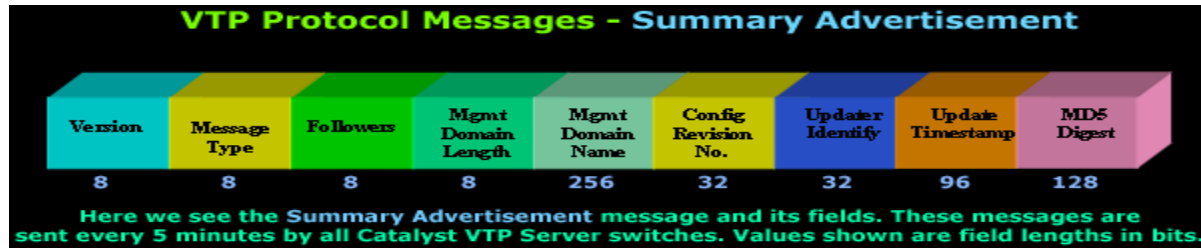
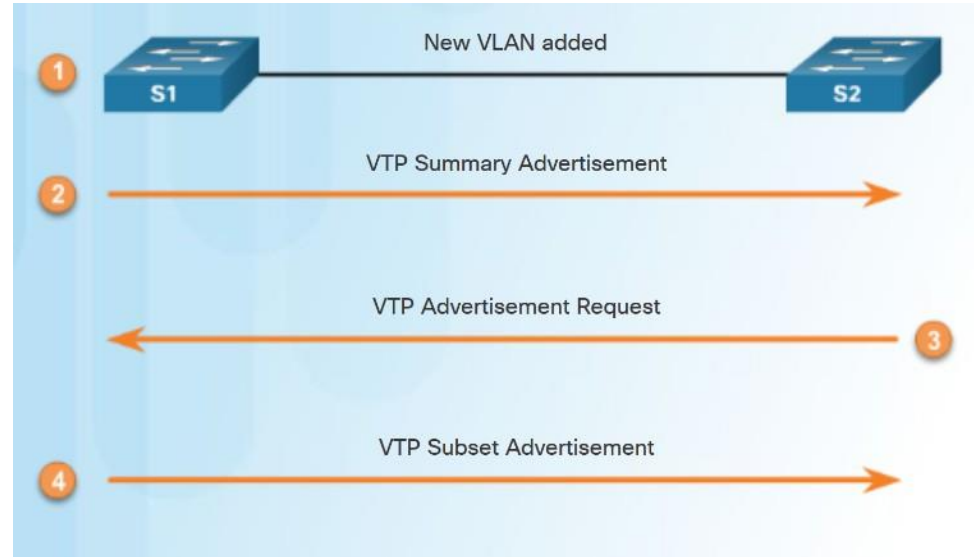


VTP Concepts and Operation

VTP Advertisements

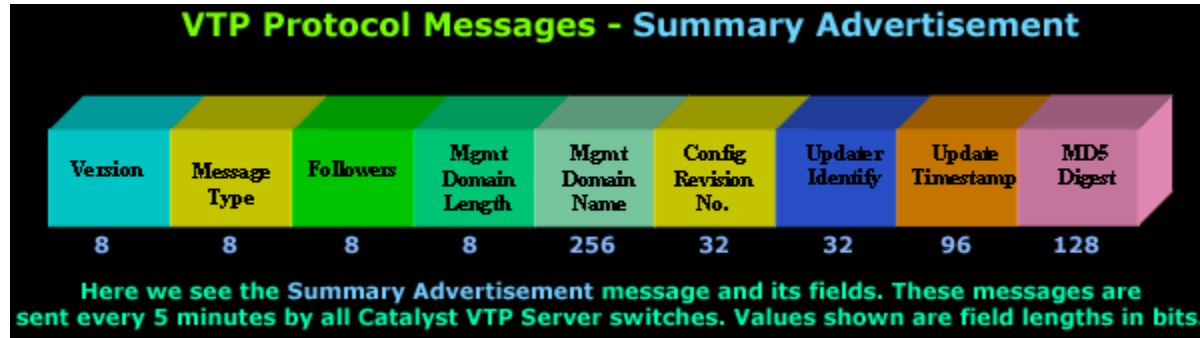
1. Summary Advertisement

VTP Servers send summary advertisements every 300 seconds and every time a VLAN database change occurs. This advertisement carries information like VTP version, domain name, revision number, number of subset advertisements to follow, time-stamp, etc. If VLAN configuration changes, one or more subset advertisements follow the summary advertisement.



VTP Revision number

The [VTP](#) revision number is a 32 bit value found in a field of the header of a VTP advertisement. **Initially, its value is zero.** The revision number is incremented by one every time there is a change in the [VLAN](#) configuration on a VTP server. A VTP client will ignore any VTP updates that have a revision number equal to or smaller than their current revision number.

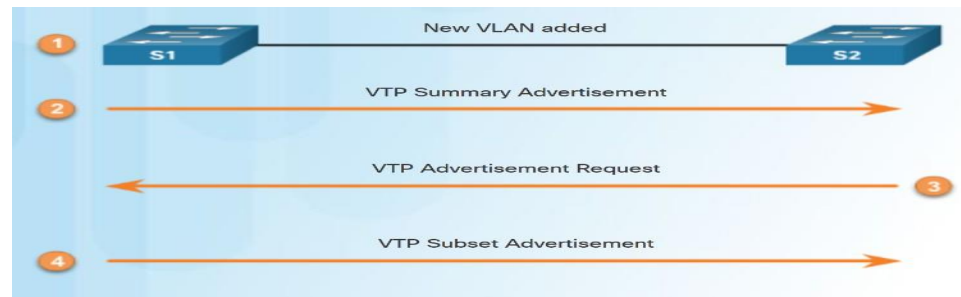


VTP Concepts and Operation

VTP Advertisements

Subset Advertisement

VTP Servers send subset advertisements after a VLAN configuration change occurs. These advertisements contain the specific changes that have been performed, like add or delete of a VLAN, change of name of a VLAN, status of a VLAN, etc.

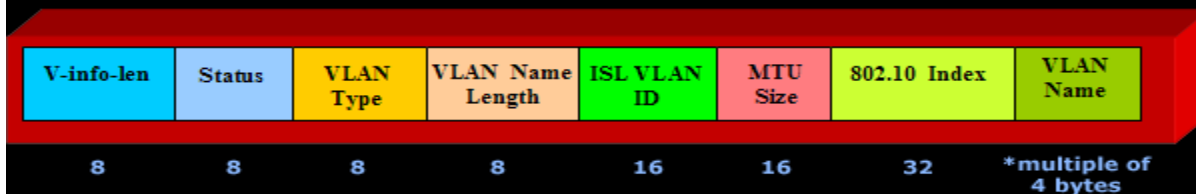


VTP Protocol Messages - Subset Advertisement



The Subset Advertisement contains a list of VLAN information. Depending on the amount of VLANs, there will be more than one Subset advertisement sent out to the network. Values shown are field lengths in bits.

VTP Protocol Messages - Subset Advertisement VLAN INFO FIELD

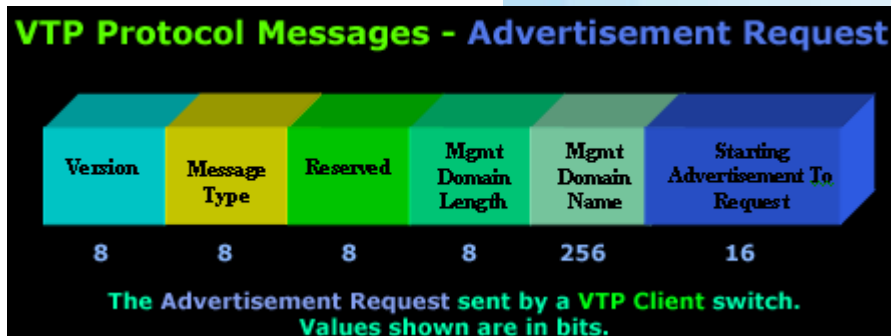
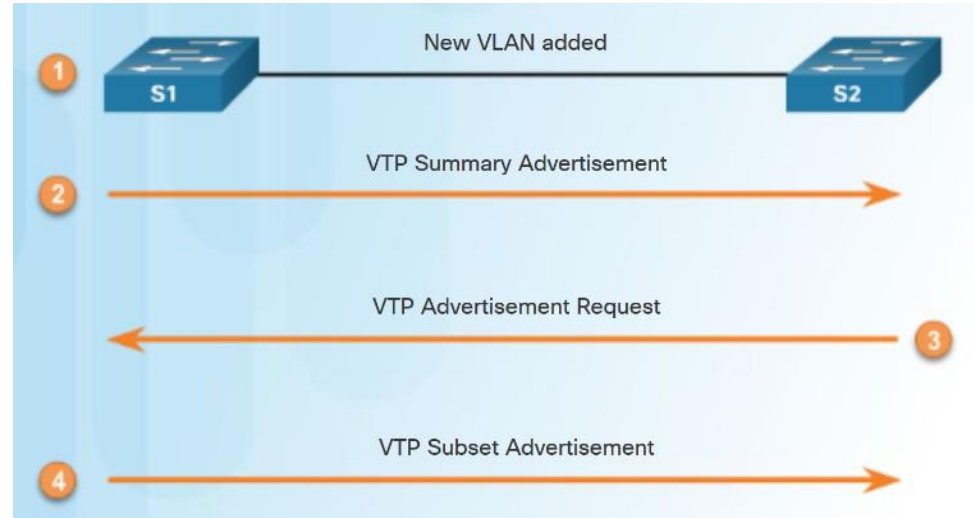


The VLAN Info Field shown here contains a further 8 subfields with all the VLAN information the network switches are expecting. Values shown are in bits.

VTP Concepts and Operation

VTP Advertisements

Advertisement Requests from Clients A VTP Client may request any VLAN information. This could be due to the Client switch might be reset and has lost VLAN information or it might hear a higher revision number than its locally stored. The Server switch responds to these Advertisement Requests with a summary advertisement followed by a subset advertisement to bring the Client up-to-date.



VTP Concepts and Operation

Default VTP configuration

Verify Default VTP Status

```
S1# show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : f078.167c.9900
Configuration last modified by 0.0.0.0 at 3-1-93 00:02:11

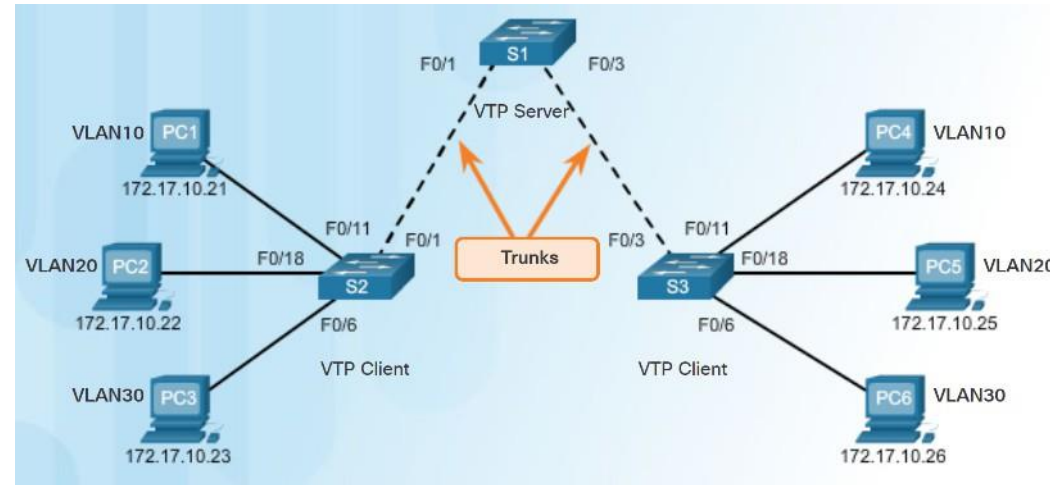
Feature VLAN:
-----
VTP Operating Mode        : Transparent
Maximum VLANs supported locally : 255
Number of existing VLANs   : 12
Configuration Revision     : 0
MD5 digest                : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
                           0x56 0x9D 0x4A 0x3E 0xA5 0x69 0x35 0xBC

S1#
```

- The **show vtp status** command displays the VTP status which includes the following:
 - VTP Version capable and running
 - VTP Domain Name
 - VTP Pruning Mode
 - VTP Traps Generation
 - Device ID
 - Configuration Last Modified
 - VTP Operating Mode
 - Maximum VLANs Supported Locally
 - Number of Existing VLANs
 - Configuration Revision
 - MD5 Digest

VTP Configuration

VTP Configuration Overview



- Steps to Configure VTP:
 - Step 1** - Configure the VTP Server
 - Step 2** - Configure the VTP Domain Name and Password
 - Step 3** - Configure the VTP Clients
 - Step 4** - Configure VLANs on the VTP Server.
 - Step 5** - Verify the VTP clients have received the new VLAN information.

VTP Configuration

Step 1 – Configure the VTP Server

```
S1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)# vtp mode ?
  client      Set the device to client mode.
  off         Set the device to off mode.
  server      Set the device to server mode.
  transparent Set the device to transparent mode.

S1(config)# vtp mode server
Setting device to VTP Server mode for VLANs.
S1(config)# end
S1#
```

```
S1# show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : f078.167c.9900
Configuration last modified by 0.0.0.0 at 3-1-93 00:02:11
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN:

```
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision    : 0
MD5 digest               : 0x57 0xCD 0x40 0x65 0x63 0x59 0x47 0xBD
                        : 0x56 0x9D 0x4A 0x3E 0xA5 0x69 0x35 0xBC

S1#
```

- Use the **vtp mode server** command to configure a switch as a VTP server.
 - Confirm all switches are configured with default configuration before issuing this command to avoid problems with configuration revision numbers.
- Use the **show vtp status** to verify.
 - Notice configuration revision number is still set to 0 and number of existing VLANs is 5.
 - The 5 VLANs are the default VLAN 1 and VLANs 1002-1005.

Step 2 – Configure the VTP Domain Name and Password

- Use the **vtp domain** *domain-name* command to configure the domain name.
 - VTP client must have same domain name as the VTP server before it will accept VTP advertisements.
- Configure a password using the **vtp password** *password* command.
 - Use the **show vtp password** command to verify.

```
S1(config)# vtp domain ?  
WORD The ascii name for the VTP administrative domain.  
  
S1(config)# vtp domain CCNA  
Changing VTP domain name from NULL to CCNA  
*Mar 1 02:55:42.768: %SW_VLAN-6-VTP_DOMAIN_NAME_CHG:  
VTP domain name changed to CCNA.  
S1(config)#
```

```
S1(config)# vtp password cisco12345  
Setting device VTP password to cisco12345  
S1(config)# end  
S1# show vtp password  
VTP Password: cisco12345  
S1#
```

Step 3 – Configure the VTP Clients

```
S2(config)# vtp mode client
Setting device to VTP Client mode for VLANs.
S2(config)# vtp domain CCNA
Changing VTP domain name from NULL to CCNA
*Mar 1 00:12:22.484: %SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to CCNA.
S2(config)# vtp password cisco12345
Setting device VTP password to cisco12345
S2(config)#
```

- Use the **vtp mode client** command to configure the VTP clients.
- Use same domain name and password as VTP server.

Step 4 – Configure VLANs on the VTP Server

- Use the **vlan** *vlan-number* command to create VLANs.
- Use **show vlan brief** to verify the VLANs.
- Use **show vtp status** to verify server status.
 - Every time a VLAN is added the configuration register is incremented

```
S1(config)# vlan 10
S1(config-vlan)# name SALES
S1(config-vlan)# vlan 20
S1(config-vlan)# name MARKETING
S1(config-vlan)# vlan 30
S1(config-vlan)# name ACCOUNTING
S1(config-vlan)# end
S1#
```

S1# show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gi0/1, Gi0/2
10	SALES	active	
20	MARKETING	active	
30	ACCOUNTING	active	
1002	fdi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fdiinet-default	act/unsup	
1005	trnet-default	act/unsup	

S1#

S1# show vtp status

```
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : CCNA
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : f078.167c.9900
Configuration last modified by 0.0.0.0 at 3-1-93 02:02:45
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN:

```
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision    : 6
MD5 digest               : 0xFE 0x8D 0x2D 0x21 0x3A 0x30 0x99 0xC8
                          : 0xDB 0x29 0xBD 0xB9 0x48 0x70 0xD6 0xB6
S1#
```

Step 5 – Verify that the VTP Clients Have Received the New VLAN Information

- Use the **show vlan brief** command to verify that the client received the new VLAN information.
- Verify client status using the **show vtp status** command.

```
S2# show vtp status
VTP Version capable      : 1 to 3
VTP Version running      : 1
VTP Domain Name          : CCNA
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : b07d.4729.2400
Configuration last modified by 0.0.0.0 at 3-1-93 02:02:45
```

Feature VLAN:

```
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs  : 8
Configuration Revision    : 6
MD5 digest               : 0xFE 0x8D 0x2D 0x21 0x3A 0x30 0x99 0xC8
                          : 0xDB 0x29 0xBD 0xE9 0x48 0x70 0xD6 0xB6
S2#
```

```
S2# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gi0/1 Gi0/2
10	SALES	active	
20	MARKETING	active	
30	ACCOUNTING	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
S2#
```


VLAN Ranges on Catalyst Switches

- Catalyst 2960 and 3560 Series switches support over 4,000 VLANs.
- Normal range VLANs are numbered 1 to 1,005.
 - Stored in vlan.dat file
- Extended range VLANs are numbered 1,006 to 4,094.
 - Not stored in vlan.dat file
 - VTP does not learn

```
Switch# show vlan brief

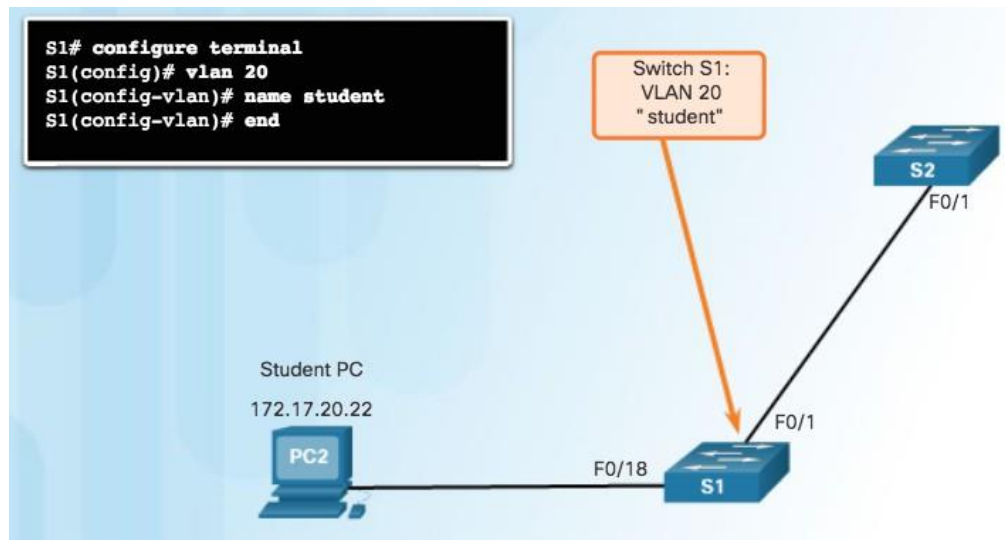
VLAN Name                Status    Ports
----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                Gi0/1, Gi0/2

1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default         act/unsup
```

Type	Definition
Normal range VLANs	• Used in small- and medium-sized business and enterprise networks. • Identified by VLAN IDs between 1 and 1005. • IDs 1 and 1002 to 1005 are automatically created and cannot be removed. (IDs 1002 through 1005 are reserved for Token Ring and Fiber Distributed Data Interface [FDDI] VLANs.) • Configurations are stored within a VLAN database file called <code>vlan.dat</code>, which is stored in flash memory.
Extended range VLANs	• Used by service providers and large organizations to extend their infrastructure to a greater number of customers. • Identified by a VLAN ID between 1006 and 4094. • Support fewer VLAN features than normal range VLANs. • Configurations are saved in the running configuration file.

Extended VLANs

Creating a VLAN



- Normal range VLANs are stored in flash in vlan.dat
- Use **vlan** *vlan-id* to create a VLAN
 - Use **name** *vlan-name* to name the VLAN
 - Naming each VLAN is considered a best practice in switch configuration.
- To configure multiple VLANs, a series of VLAN IDs can be entered separated by commas, or a range of VLAN IDs separated by hyphens.
 - **vlan 100,102,105-107**

Assigning Ports to VLANs

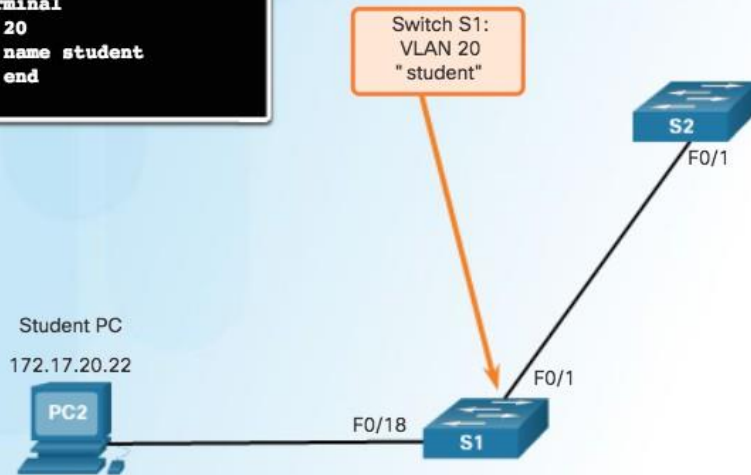
- Access port can belong to only one VLAN at a time.
 - Only exception is when an IP phone is connected to the port. Then there are two VLANs associated with the port: one for voice and one for data.

Note: Use the **interface range** command to simultaneously configure multiple interfaces.

Cisco Switch IOS Commands

Enter global configuration mode.	S1# configure terminal
Enter interface configuration mode.	S1(config)# interface interface_id
Set the port to access mode.	S1(config-if)# switchport mode access
Assign the port to a VLAN.	S1(config-if)# switchport access vlan vlan_id
Return to the privileged EXEC mode.	S1(config-if)# end

```
S1# configure terminal
S1(config)# vlan 20
S1(config-vlan)# name student
S1(config-vlan)# end
```



Verifying VLAN Information

```
S1# show vlan name student
```

VLAN	Name	Status	Ports
20	student	active	Fa0/11, Fa0/18

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
20	enet	100020	1500	-	-	-	-	-	0	0

```
Remote SPAN VLAN
```

```
Disabled
```

Primary	Secondary	Type	Ports

```
S1# show vlan summary
```

```
Number of existing VLANs      : 7
Number of existing VTP VLANs  : 7
Number of existing extended VLANs : 0
```

```
S1#
```

Commands to verify VLANs:

- **show vlan**
- **show interfaces**
- **show vlan name *vlan-name***
- **show vlan brief**
- **show vlan summary**
- **show interfaces vlan *vlan-id***

Configuring Extended VLANs

```

S1(config)# vtp mode transparent
Setting device to VTP Transparent mode for VLANs.
S1(config)# vlan 2000
S1(config-vlan)# end
S1#

```

```

S1# show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gi0/1, Gi0/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	
2000	VLAN2000	active	

```

S1#

```

- Extended range VLANs are identified by a VLAN ID between 1006 and 4094.
- To configure an extended VLAN on a 2960 switch it must be set to VTP transparent mode. (By default 2960 switches do not support Extended range VLANs.)

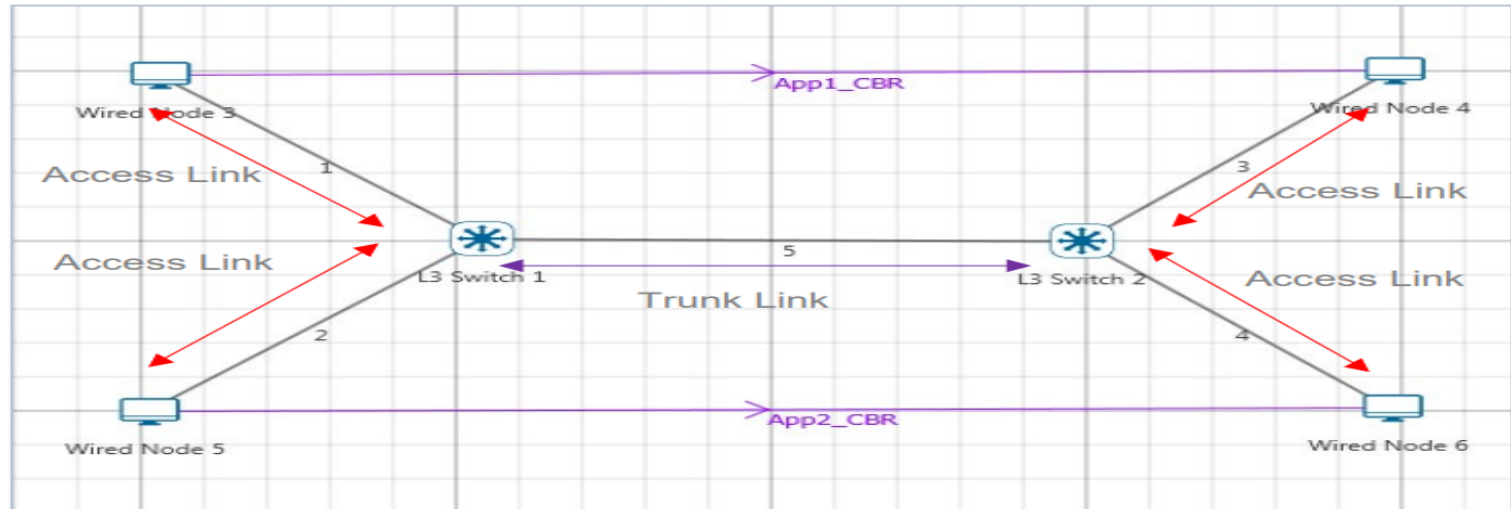
DTP

(Dynamic Trunk Protocol)

Trunk and Access Links

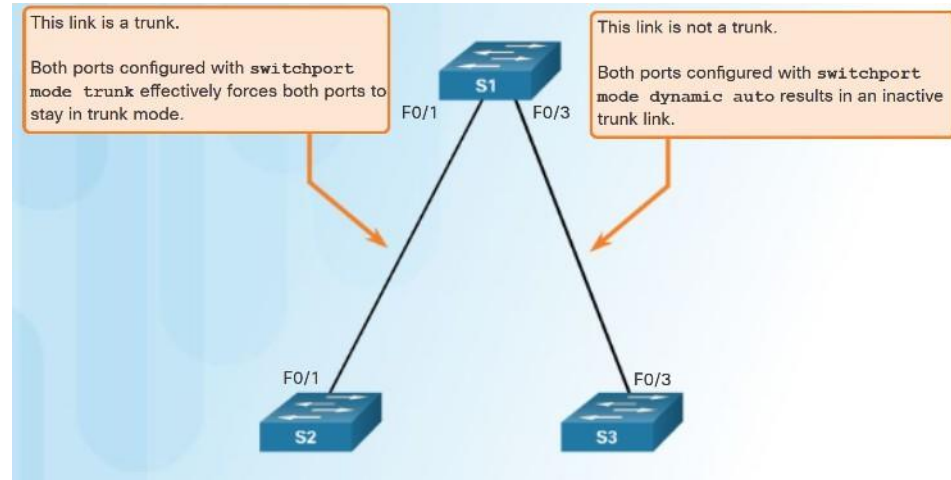
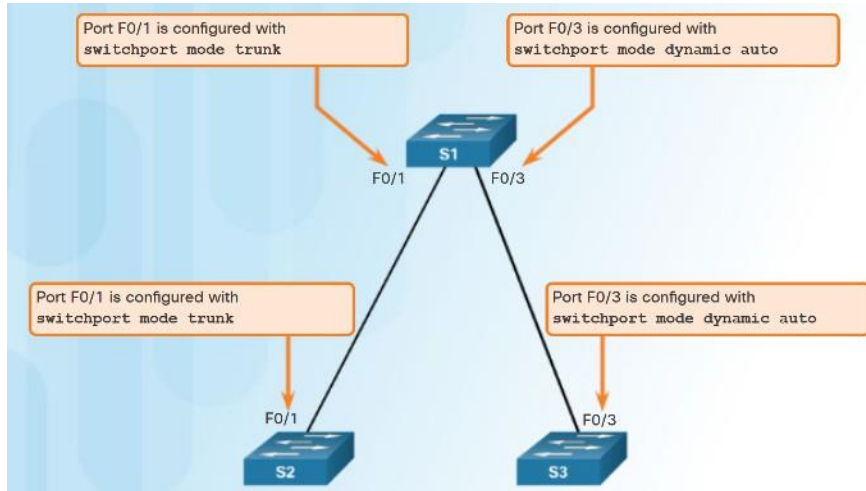
- **Trunk links** are physical links between switches or between switches and routers. The data frames transmitted on a trunk link must be tagged so that devices can identify the VLAN to which the data frames belong. Therefore, a trunk link can carry data frames of multiple VLANs.
- **Access links** are the links connecting the end devices. These are the links usually carrying the Data VLAN information. The link between the switches is called trunk link. It carries packets from all the VLANs

An access link **carries the traffic of a single VLAN**, whereas a trunk link connection carries the traffic of multiple VLANs



Dynamic Trunking Protocol

Introduction to DTP



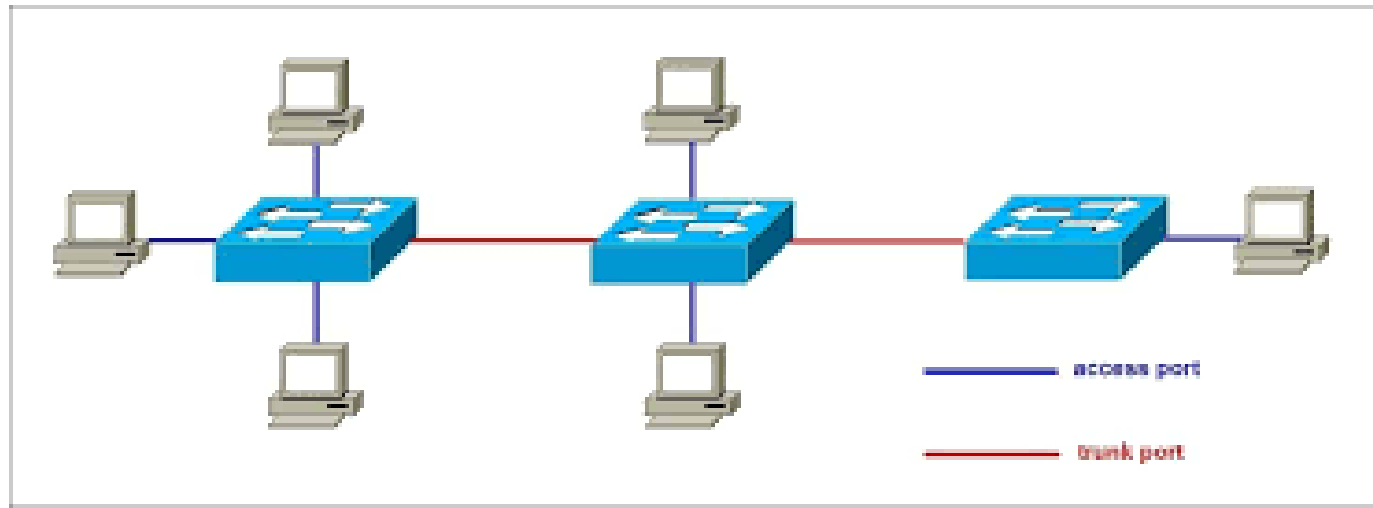
- Trunk negotiation is managed by the Dynamic Trunking Protocol (DTP)
 - DTP is a Cisco proprietary protocol
 - automatically enabled on Catalyst 2960 and Catalyst 3560 Series switches.
- To enable trunking from a Cisco switch to a device that does not support DTP, use the **switchport mode trunk** and **switchport nonegotiate**

Dynamic Trunking Protocol Introduction to DTP

The DTP is used by Cisco switches to negotiate whether an interconnection between two switches should be put into access or trunk mode. It is meant both to ease the initial deployment of a switched network and to minimize configuration errors that result from mismatched port configuration on an interconnection between two switches.

The DTP helps to automatically negotiate whether the port should be put into access or trunk mode and what trunking protocol (802.1Q or ISL) should be used. The individual DTP modes are:

- dynamic auto - the port will negotiate the mode automatically, however, it prefers to be an access port
- dynamic desirable - the port will negotiate the mode automatically, however, it prefers to be a trunk port



Dynamic Trunking Protocol

Negotiated Interface Modes

- Different trunking modes:
 - **Switchport mode access** - interface becomes a nontrunk interface.
 - **Switchport mode dynamic auto** - interface becomes a trunk if the neighboring interface is set to trunk or desirable mode.
 - **Switchport mode dynamic desirable** - interface becomes a trunk if the neighboring interface is set to trunk, desirable, or dynamic auto mode.
 - **Switchport mode trunk** - interface becomes a trunk even if the neighboring interface is not a trunk interface.
 - **Switchport nonegotiate** - prevents the interface from generating DTP frames.

```
S1# show dtp interface f0/1
DTP information for FastEthernet0/1:
  TOS/TAS/TNS:                TRUNK/ON/TRUNK
  TOT/TAT/TNT:                802.1Q/802.1Q/802.1Q
  Neighbor address 1:          0CD996D23F81
  Neighbor address 2:          000000000000
  Hello timer expiration (sec/state): 12/RUNNING
  Access timer expiration (sec/state): never/STOPPED
  Negotiation timer expiration (sec/state): never/STOPPED
  Multidrop timer expiration (sec/state): never/STOPPED
  FSM state:                   S6:TRUNK
  # times multi & trunk        0
  Enabled:                     sim
  In STP:                       no

<output omitted>
```

- Configure trunk links statically whenever possible.
- Use **show dtp interface** to verify DTP.

DTP datagrams are also sent if the port is set statically to the trunk mode. However, if the port is set statically to the access mode, both sending and processing DTP datagrams on that port is deactivated.

The individual combinations of port settings lead to following results:

- dynamic auto + dynamic auto = access
- dynamic auto + dynamic desirable = trunk
- dynamic desirable + dynamic desirable = trunk
- dynamic auto or dynamic desirable + trunk = trunk
- dynamic auto or dynamic desirable + access = access

As you can see, if both ports are dynamic auto, they will act as access ports. If either of them is dynamic desirable, both will agree on trunking. If one of them is dynamic and the other is static, the mode is dictated by the statically set port.

DTP & Trunk Modes

